

PalmSource®

Conference and Expo



Get the Technical Scoop on Handspring's Latest Products

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Agenda

- Treo
  - Overview
  - Hardware
  - Software
- Opportunities
- Application Development
  - 3rd party SMS Application
  - SIM Toolkit
  - Phone Application
  - Data Connection
  - Keyboard Aware application





Treo Hardware



Treo Software



PHONEBOOK

DATEBOOK

BROWSING

MESSAGING









Phonebook







- Instant lookup
  - One-handed with keyboard
  - 3-ways to access
    - Initials
    - Last name
    - First name
- 1,000 entry detailed call history
- Easy 3-way conference calling
- Caller ID, know who's calling by name
- Personal speakerphone mode
- Speed dial
- Dial pad
- Customizable ring tones

Palm OS® Organizer

- Datebook+
- Compatible with 11,000+ Palm OS apps
  - Wireless solutions
  - Enterprise solutions
  - Multimedia
  - Transferring Data
  - IR, etc…









Blazer 2.0



- Intuitive graphical user interface
- Access to revenue-generating carrier services
- Support for multiple Web and wireless standards
  - HTML, WML, HDML, cHTML (i-mode), xHTML
- Support for strong SSL encryption (128 bit)
- Designed for Palm OS handheld computers
  Optimized for device: screen size, processing power, color support, text entry, etc.
- Fast, pages begin loading in seconds

Wireless Messaging



SMS Messaging



POP3 e-mail



Automated e-mail

So what are the opportunities?





Working with wireless operators



mmO2

SMARTONE

VoiceStream
Global Wireless by T-Mobile

SingTel

TELIA

Wireless Application development

- 3rd party SMS Application
- SIM Toolkit
- Phone Application:
  - Call Logging Database
  - Ring Tone
  - How to Dial a Phone #
- Data Connection
- Other
  - Keyboard Aware application including the Rocker Switch
  - Lid Control
  - Power/Ringer switch

SMS Application…

- Use the GSM and SMS Library to:
  - Send, Receive and Encode Text Messages (SMS character encoding/decoding)
  - Manage SMS messages
- How to use the library?
  - Load the library:
    - GSM Library Name: "GSMLibrary.lib“
    - GSM DB Library Name: "GSM Library“
    - GSM Library CreatorID: 'GSM!'
  - Register with the library
  - Un-register with the library when you’re done!



…SMS Application: Initialization…

- Register with the library
  - GSMLibRegister
  - GSMLibSetDataApplication (VisorPhone only)
- Accessing the Message Database
  - GSMLibGetDBRef
  - GSMLibReleaseDBRef
- Events:
  - kMessageInd
  - kSegmentInd
  - kMessageStat
  - kMessageDel
  - kMessageMoved
  - kError
  - kButton (VisorPhone only)
- Miscellaneous
  - GSMLibSetModulePower
  - GSMLibModulePowered
  - GSMLibRegistered
  - GSMLibRoaming

…SMS Application: Message Database…

- Message Management
  - GSMLibNewMessage
  - GSMLibSendMessage
  - GSMLibDeleteMessage
    - If you are the owner
- Message Types
  - kMTIncoming
  - kMTOutgoing
- Message Status (Category)
  - kNone
  - kReceiving
  - kReceived
  - kPending
  - kSending
  - kSent



…SMS Application: Message Content

- Setting Message Data
  - GSMLibSetText
  - GSMLibSetDate
  - GSMLibSetOptions
  - GSMLibSetAddresses
  - GSMLibSetStatus
  - GSMLibSetFlags
  - GSMLibSetOwner
- Getting Message Data
  - GSMLibGetText
  - GSMLibGetDate
  - GSMLibGetOptions
  - GSMLibGetAddresses
  - GSMLibGetStatus
  - GSMLibGetFlags
  - GSMLibGetOwner
  - GSMLibGetType



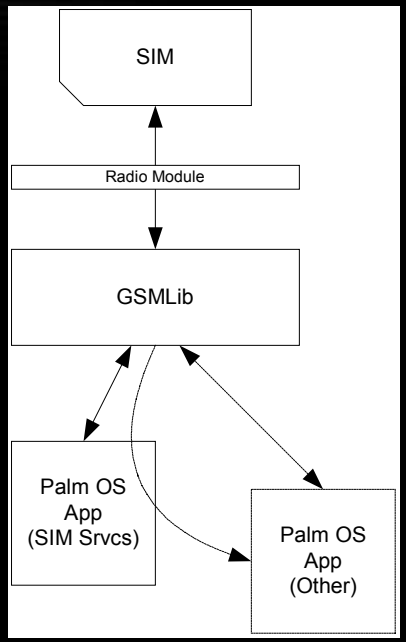
…SMS Application: Addresses …

- Manipulating Characters
  - GSMLibIsLegalCharacter
  - GSMLibMapCharacter
  - GSMLibGetSubstitution
  - GSMLength
- Preferences
  - GSMLibSetServiceCentreAddress
  - GSMLibGetServiceCentreAddress
- Managing Addresses & Lists
  - GSMLibNewAddress
  - GSMLibSetField
  - GSMLibGetField
  - GSMLibNewAddressList
  - GSMLibAddAddress
  - GSMLibDisposeAddressList
  - GSMLibGetNth
  - GSMLibSetNth
  - GSMLibCount

…SMS Application

- SMS Library
  - You need to get the following:
    - “SMSLibraryHeaderFiles_101.exe”
      - http://www.handspring.com/developers/sw_dev.jhtml
    - Application note: “AN-18: Using the SMS Library (1.00)”
      - http://www.handspring.com/developers/tech_notes.jhtml
    - Try the SMS Simple Sample code
  - If you are using VisorPhone, you also need to update the VisorPhone software:
    - http://www.handspring.com/developers/wireless/index.jhtml
    - Enable 3rd party application to receive SMS messages

SIM Toolkit…



- We have support for:
  - SIM Initiated commands
  - User Initiated commands
    - Palm OS® app
- 3rd party application
  - There is a default built-in App
    - “SIM Srvcs”
  - However, you can register with the GSM Library
    - Notifications will be sent via standard PalmOS sub - launch method and events
- Details available on the web:
  - Treo Product Technical Specification & Development Guide

… SIM Toolkit

- There are 3 functions
  - `GSMLibSATGetMainMenu`
  - `GSMLibSATEndSession`
  - `GSMLibSATSendRequest`
- What you should do to use them:
  - register itself to the GSM Library
  - start an SAT session with a call to `GSMLibSATGetMainMenu`
    - App will receive the main menu
  - Depending on the user interaction, app can call `GSMLibSATSendRequest`
    - `GSMLibSATSendRequest` is a one-way communication
  - App will receive information back from the GSMLib as a GSM Event.
  - Anytime the app can end the current session by using `GSMLibSATEndSession`

Phone Application: Dial a Phone number

- Use the Address Book Helper
  - Uses the Palm OS Notification Manager:
  - Implements a set of three action codes:
    - Enumerate
    - Validate
    - Execute
  - Service Class:
    - Voice: Service Class to make a voice telephone call

Phone Application: Call Logging Database…

- Normal Palm OS® Database
  - Type: “call”
  - Creator ID: “HsPh”
- Record structure: `CallHistoryRecordType`

```
typedef struct {
  UInt16   flags;
  DateType date;              /*Date of the call*/
  TimeType time;              /*Time of the call*/
  UInt32   duration;          /*Duration in seconds*/
  char*    name;              /*Name of caller or NULL*/
  char*    number;            /*Phone Number of caller or NULL*/
} CallHistoryRecordType;
```

…Phone Application: Call Logging Database

- Record Categories:
  - #define callCategoryIncoming 1
  - #define callCategoryMissed 2
  - #define callCategoryOutgoing 3

- See Application Note:
  - “AN-22: How to Use the Call History Database” http://www.handspring.com/developers/tech_notes.jhtml

Ring Tone…

- Palm OS MIDI File Format
- System Ring Tone Database information:
  - name: “System Ring Tones”
  - Type: “smfr”
  - creator ID: “GSMr”
- Optionally:
  - New “public” creator ID/Type for Ring Tone Databases:
    - Creator ID: “HSsf”
    - Type: “smfr”
  - Applications can search for all Ring Tones in system
  - Database name should include your own Creator ID

… Ring Tone

- New API that you can call to set Ring Tone:
  - GSMLibGetRingingInfo
  - GSMLibSetRingingInfo
  - GSMLibGetToneIDs
  - GSMLibGetToneName
  - GSMLibPlayTone
  - GSMLibStopTone
  - GSMLibGetSMSRingInfo
  - GSMLibSetSMSRingInfo

Available end of February in new Handspring header files



Data connection: EASY!

- Same as any standard Palm OS 3.5 handheld!
  - New connection entry: Wireless Modem

Others: Keyboard Aware Application

- Rocker Switch
  - Events that looks like a keyDownEvent
    - Jog Up ⇒ vchrPageDown
    - Jog Down ⇒ vchrPageUp
    - Jog Press ⇒ hsChrJogScan
- Keyboard:
  - Option button with lock (aka shift key)
  - New functions in HsExt.h to allow detection of many more keys being pressed:
    - KeyCurrentState() ⇒ HsKeyCurrentStateExt()
    - KeySetMask() ⇒ HsKeySetMaskExt()
    - GrfSetState() ⇒ HsGrfSetStateExt()
    - GrfGetState() ⇒ HsGrfGetStateExt()



…Keyboard

- Conversion of key character to their “Option” value and vice-versa:
  - HsKeyRegChrToOptChr
  - HsKeyOptChrToRegChr



Lid Control

- HsPrefSet (

  hsPrefLidStatusChgCharAppCreator,

  kYourAppCreator,

  kYourAppCreatorSize)

- Get a keyDownEvent with the vchrPowerOff character when the lid is closed



Power and Ringer Switch

- Power Switch
- Ringer Switch



Q & A

- Register as a developer

  http://www.handspring.com/developers/index.jhtml

- Download the Treo Product Development Guide

  http://www.handspring.com/developers/documentation.jhtml



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